

Vasil Topalovic

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EDUCATION

University Of Ottawa

Co-op Honours Bachelor of Computer Science - 9.2/10 CGPA (3.68 GPA)

Ottawa, ON

Sept 2022 - Present

EXPERIENCE

Software Engineer

Ciena

Sept 2025 - Present

Ottawa, ON

- Designed a distributed task execution platform using Python, Celery, and RabbitMQ, increasing throughput by 50%.
- Developed a Dagster pipeline to extract and process user-uploaded logs, increasing adoption to 1 additional product team.
- Automated deployments using Ansible and Docker, reducing manual workload by 90%.
- Developed an internal Vue.js + Python log analysis tool, improving root cause analysis and cutting debugging time by 5%.

Software Engineer (Freelance)

IgNITE Medical Case Competition

Sept 2024 - January 2025

Ottawa, ON

- Created a streamlined meeting booking system using SvelteKit and Rust, reducing the admin workload by 2 hours/week.
- Integrated Stripe to securely manage payments for educational videos and PDF resources, enabling 200+ participants to access educational material.

Data Engineer

Canada Border Services Agency

May 2024 - Dec 2024

Ottawa, ON

- Implemented, tested, and deployed a pipeline to ingest data from AWS S3 Buckets to Azure Storage Containers in Azure Data Factory, reducing storage costs by 10%.
- Performed and maintained data validation, identifying and resolving discrepancies in 0.2% of database records.

PROJECTS

Web Search Engine | React, TypeScript, Rust, PostgreSQL, Redis

- Engineered a multi-threaded web crawler and indexer in Rust, utilizing PostgreSQL for persistent data storage and Redis to manage concurrent task queues, indexing a total of 1,000,000+ pages.
- Developed a responsive React front-end integrated with the core search API, delivering sub-300ms search queries.

Online Multiplayer Video Game | TypeScript, PostgreSQL, Godot

- Shipped an OpenAPI-documented Hono + Drizzle/Postgres REST API with Steam auth, serving 1,200+ players.
- Designed a compact binary replay format that supports moderated review of leaderboard submissions.
- Built paginated per-level leaderboards, Discord integrated world-record notifications, and user-hosted server discovery.

Programming Language Compiler | C++, LLVM

- Built a recursive-descent parser capable of processing arrays, structs, pointers, and arithmetic/boolean expressions.
- Implemented static type checking and a multi-stage lowering pipeline with a custom intermediate representation.
- Integrated the compiler with the LLVM toolchain to generate optimized machine code for multiple architectures.

ACHIEVEMENTS

1st Place: HackTheHill II Hackathon.

1st Place: uOttHack 7 Hackathon.

1st Place: NASA Space Apps Hackathon. (Ottawa Region)

TECHNICAL SKILLS

Languages: C/C++, Rust, JavaScript/TypeScript, Java, Python, SQL, HTML/CSS

Frameworks / Libraries: Vue, SvelteKit, React, Node.js, Tailwind, Qt, Dagster, FastAPI, Express, wgpu, LLVM

Developer Tools: Git, Docker, Linux, AWS, Azure, PostgreSQL, MongoDB, Redis, Firebase, Jira, Confluence